

PROFESSIONAL SUMMARY

Senior **Python/Flask** Backend engineer with 12+ years across SaaS, property technology, nonprofit platforms, civic technology, and IT services, specializing in **Python 3.10-3.12**, **Flask**, **SQLAlchemy**, **PostgreSQL**, **Redis**, **AWS** for secure, scalable platforms; delivers **architecture** ownership, modernization, **performance** tuning, **test automation**, **CI/CD** discipline, and **production reliability** that improves customer workflows, **release quality**, and engineering velocity.

TECH SKILLS

- Backend Engineering: **Python 3.10-3.12**, **Flask**, **SQLAlchemy**, **PostgreSQL**, **Redis**, **AWS**, service-layer design, API gateways, **OpenAPI**, **REST APIs**, **GraphQL**, **gRPC**, async processing, background jobs
- Databases & Messaging: **PostgreSQL** 13-16, **MySQL** 8, **SQL Server**, MongoDB 6-7, **Redis** 6-7, **Kafka**, **SQS**, **RabbitMQ**, indexing, query tuning
- Cloud & DevOps: **AWS**, **Azure**, **GCP**, **Docker**, **Kubernetes**, **Terraform** 1.x, **GitHub Actions**, **Jenkins**, secrets management, deployment automation
- Testing & Reliability: Unit testing, integration testing, contract testing, load testing, mutation-safe migrations, **observability**, **incident response**
- Senior Practices: System design, **microservices**, **event-driven architecture**, domain-driven design, **security** reviews, mentoring, production ownership

PROFESSIONAL EXPERIENCE

Staff Software Engineer

Relatio

Jun 2022 - Apr 2026

Las Vegas, NV

- Led backend **architecture** for a SaaS workflow and analytics platform using **Python 3.10-3.12**, **Flask**, **SQLAlchemy**, **PostgreSQL**, **Redis**, **AWS**, improving API response reliability by **28%** through clearer service boundaries and async processing.
- Designed versioned **REST** and **GraphQL** APIs with strong contracts, validation, and error semantics, reducing frontend-backend integration defects by **19%** across releases.
- Migrated legacy synchronous services into containerized **microservices** with repeatable builds, staged deployment gates, and production-safe environment validation.
- Resolved recurring production errors caused by slow ORM queries, missing composite indexes, inconsistent pagination, and non-idempotent background retries.
- Implemented queue-based processing, scheduled jobs, and retry-safe workflows to stabilize report generation, customer notifications, and data synchronization under peak load.
- Created **observability** standards with structured logging, tracing metadata, dashboards, and alert thresholds, improving incident diagnosis speed by **31%** during support rotations.
- Refactored authentication and authorization checks into reusable middleware patterns that strengthened API **security** and reduced duplicated permission logic across services.
- Partnered with product managers to implement configurable workflow rules, audit trails, reporting exports, and scheduled processing for enterprise customers.
- Guided engineers on service design, database migration strategy, code review discipline, and production debugging so new backend services stayed maintainable.

Senior Software Engineer

AppFolio

Feb 2018 - May 2022

Goleta, CA

- Built backend services for property technology products using earlier versions of **Python 3.10-3.12**, **Flask**, **SQLAlchemy**, **PostgreSQL**, **Redis**, **AWS**, improving rental workflow processing speed by **24%** through API and database optimization.
- Implemented tenant communication, payment-related APIs, document workflows, and background processing with secure validation and transaction logging.
- Migrated older backend modules from tightly coupled views and scripts into service-oriented patterns with stronger tests and reusable business rules.
- Fixed production bugs involving duplicate payment notifications, timeout errors, inconsistent status transitions, and missing retry safeguards by adding idempotency controls.
- Optimized relational queries for search, lease records, reporting pages, and high-read lookup data by adding indexes and removing expensive N+1 access patterns.
- Developed asynchronous job workflows for document generation, email events, and scheduled customer notifications with clear retry policies and operational visibility.
- Improved API validation with schema checks, reusable error responses, and regression tests that made backend behavior more predictable for frontend and mobile teams.
- Mentored mid-level developers on backend design, testing strategy, database modeling, debugging, and production support during feature delivery.

Software Engineer

Blackbaud

Jul 2016 - Jan 2018

Charleston, SC

- Developed backend features for nonprofit fundraising and donor management systems using service APIs, relational data models, and message-based processing.
- Improved donor search and reporting workflows by tuning database queries and API response structures, reducing common reporting delays by **21%** for large datasets.
- Implemented secure data import services with validation rules, duplicate detection, transactional processing, and detailed error reporting for support teams.

- Resolved bugs related to malformed CSV imports, timezone conversion issues, partial transaction failures, and inconsistent campaign totals by adding stricter parsing.
- Built reusable API components for campaign records, donor profiles, activity history, and reporting flows with consistent backend behavior.
- Added automated test coverage around donation, reporting, and import workflows, improving release confidence and reducing regression defects before production.
- Worked with DevOps to improve deployment checks, environment configuration, and application logging so production issues could be identified faster.

Software Developer

Granicus

May 2014 - Jun 2016

Dallas, TX

- Built civic technology backend features for agenda management, document publishing, public records, and subscription workflows used by government-facing platforms.
- Improved public meeting content processing and notification reliability through better job scheduling, retry handling, backend validation, and operational logging.
- Implemented API endpoints for agenda management, document metadata, and subscription workflows while keeping permission checks consistent across internal modules.
- Fixed recurring production issues involving failed document uploads, missing metadata, slow search responses, and incomplete notifications by improving validation and indexing.
- Migrated older utility scripts into maintainable backend services with clearer configuration management, logging, and reusable exception-handling patterns.
- Worked with support teams to investigate customer incidents, reproduce backend errors, and deliver safe patches without disrupting active government portals.

Junior Software Developer

Vology

Jun 2013 - Apr 2014

Gainesville, FL

- Supported internal IT services applications using backend scripts, database utilities, shell automation, **MySQL**, and Linux tools for maintenance workflows.
- Built small automation services for ticket routing, inventory updates, and operational reporting that reduced repetitive manual work for internal teams.
- Fixed bugs in legacy scripts involving bad input handling, missing null checks, inconsistent database updates, and unclear failure messages by adding safer validation.
- Assisted senior engineers with API maintenance, database scripts, and deployment preparation while learning production debugging and software delivery practices.
- Wrote unit tests, technical notes, and troubleshooting guides for backend utilities, helping the team maintain older systems with fewer support escalations.

SELECTED PROJECTS

- High-Volume API Modernization: Re-architected backend services with **Python 3.10-3.12**, **Flask**, **SQLAlchemy**, **PostgreSQL**, **Redis**, **AWS**, **PostgreSQL**, **Redis**, **Docker**, and **AWS** to improve request reliability, reduce timeout errors, and support safer schema migrations.
- Event Processing and Reporting Engine: Built asynchronous jobs, queue consumers, retry policies, and audit-friendly data flows with **Python 3.10-3.12**, **Flask**, **SQLAlchemy**, **PostgreSQL**, **Redis**, **AWS**, helping operations teams process reports and notifications with stronger production visibility.

EDUCATION

Arkansas State University - Bachelor's Degree in Computer Science, Sep 2009 - May 2013